

KARTHIK RENGARAJAN

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PROFESSIONAL SUMMARY

Accomplished Senior Controls Engineer and Doctoral Candidate specializing in Autonomous Systems and Robotics as the core focus of D.Eng. research and professional career. Extensive expertise in vehicle dynamics, non-linear adaptive control strategies, motion control, and functional safety (ISO 26262) for autonomous and electrified vehicle platforms. Proven track record of developing novel control algorithms, authoring multiple utility patent applications cited by major automotive OEMs, and contributing as a peer reviewer for premier control system conferences.

EXPERIENCE

Nuro | Senior Controls Engineer | Feb 2026 – Present

- Lead software development for motion controllers and trajectory processors on the Uber Nuro Lucid Robotaxi program.
- Implement Safety Requirement Specifications (SSRs) complying with ISO 26262 functional safety standards.
- Drive advanced research initiatives in controls state estimation for next-generation autonomous platforms.

Nexteer Automotive, Auburn Hills, MI | System Controls Engineer II | Aug 2022 – Feb 2026

- Led full-cycle development of critical Steer-by-Wire (SbW) features, including End of Travel Management, Misalignment Management, ADAS integration, and Vehicle Disturbance Rejection.
- Engineered and validated core SbW control algorithms through comprehensive MIL, HIL, and in-vehicle testing.
- Designed advanced chassis control strategies using nonlinear sliding mode control and backstepping in CarSim-MATLAB co-simulations.
- Enhanced ADAS performance with novel algorithms mitigating Torque Steer and Brake Pull while dynamically adjusting driver feel.
- Designed safety-critical Pinch Detection algorithms for electronically stowable steering columns.
- Executed software integration and builds for prototype deployment in AUTOSAR using Vector DaVinci and hand-coded C.

TuSimple, Tucson, AZ | Vehicle Controls Engineer (Integration) | Feb 2022 – Aug 2022

- Managed longitudinal control, power management, and power bridge controls for autonomous heavy-duty trucks.

FEV North America, Inc., Auburn Hills, MI | Control Systems Engineer (Electrified Powertrain) | Feb 2019 – Feb 2022

- Developed Torque Management Control Strategies for hybrid systems using mathematical modeling in Agile workflows.
- Designed and optimized Motor Speed PID Control modules governing engine and clutch slip speed for PHEV/MHEV programs.
- Served as technical focal point for Engine Start-Stop Controls, aligning Systems and Calibration teams.
- Authored DV test cases, defined acceptance criteria, and executed DVP&R testing across MIL/SIL/HIL platforms using dSPACE and ETAS-INCA.
- Participated in system and subsystem DFMEA activities during requirements development.

EDUCATION

Doctor of Engineering (D.Eng.) in Computer Engineering / Electrical & Computer Engineering

University of Michigan, Dearborn | Jan 2025 – Present

- Doctoral Candidate under [Dr. Alireza Mohammadi](#) with research focuses on advanced control strategies for autonomous sailboat motion planning and control using nonlinear controls and Reinforcement Learning.

Master of Engineering in Mechanical Engineering (Specialization: Controls)

Texas A&M University, College Station, TX | Dec 2018 | GPA: 3.88 / 4.0

Bachelor of Technology in Mechanical Engineering

Amrita University, India | Jun 2016 | GPA: 9.38 / 10.0

SCHOLARLY & PROFESSIONAL CONTRIBUTIONS

- Patent Portfolio: Author of 9 Utility Patent Applications (3 granted, 1 published, 1 provisional application).
- Industry Citations: Patents cited by leading global automotive and technology companies, including General Motors: <https://patents.google.com/patent/US12570351B2/en>, Honda, and Samsung.
- Invited Peer Reviewer: Modeling, Estimation, and Control Conference (MECC 2026) — Evaluated original research manuscripts on data-driven safety verification and gain synthesis for autonomous systems.
- Conference Paper: Co-author on accepted conference abstract regarding advanced control systems (Upcoming full manuscript review before presentation).
- Journal Research: Manuscript in preparation for submission to an Elsevier peer-reviewed journal.
- Open Research Report (Zenodo / CERN): "[Condition Monitoring of Motorcycle Front Suspension Using Vibration Analysis and Machine Learning](#)" (Published May 2026, Version v1; research conducted in 2016). Highlighted pioneering application of Machine

Learning techniques in vibration analysis, predating the current widespread industry adoption of ML in this field.

CERTIFICATIONS & ACADEMIC TRAINING

- Automotive Functional Safety Professional Certification: ISO 26262 (SGS TUV) (2021)
- Algorithms for Battery Management Systems: Coursera (Focus on SOC, SOH, and Li-Ion battery parameter estimation) (2020)
- Fundamentals of Vehicle Dynamics: SAE (2025)
- C++ Nanodegree: Udacity (Focus on Object-Oriented Programming, Memory Management, and Concurrency) (2022)
- Self-Driving Car Engineer Nanodegree: Udacity (2021)

TECHNICAL SKILLS & EXPERTISE

- **Core Domain Expertise:** Autonomous Systems, Autonomous Vehicles, Robot Motion Control, Vehicle Dynamics, Nonlinear Adaptive Controls (Sliding Mode, Backstepping), Control System Theory, State Estimation (Kalman Filters, Particle Filters), Hybrid and Electric Powertrains.
- **Systems & Safety Engineering:** Functional Safety (ISO 26262), ASPICE, AGILE Methodologies, Machine Learning, Deep Neural Networks.
- **Tools & Platforms:** MATLAB/Simulink/Stateflow, CarSim, dSPACE-ControlDesk, ETAS-INCA, Vector CANoe, CANape, AUTOSAR, Vector DaVinci Developer & Configurator, IBM DOORS, IBM Jazz, GIT.
- **Languages & Protocols:** Python, C++, C, CAN/LIN protocols.